

Ryan Hassell

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PROFESSIONAL EXPERIENCE

Associate Software Engineer - Advanced Artificial Intelligence
Cobbs Creek Healthcare

June 2025 - Present
Newtown Square, PA

- Developed a production-grade AI analytics chatbot supporting natural language querying over 10+ enterprise datasets, generating 25+ visualization types with contextual memory.
- Built multi-agent AI workflows using LangChain and LangFlow for retrieval, SQL planning, visualization selection, and explanation across end-to-end analytics pipelines.
- Designed LLM-driven literature review generators and interactive analytics dashboards backed by AWS, Azure, Snowflake, and Databricks.
- Collaborated with clinicians and data scientists to integrate agentic AI systems into live research and analytics workflows.

Tech Stack: Python, TypeScript, JavaScript, SQL, LangChain, LangFlow, FastAPI, React, Streamlit, R Shiny, Plotly, AWS, Azure, Snowflake, Databricks

Software Engineer Intern
Insomniac Design

June 2023 - June 2024
Washington, D.C.

- Led backend user and data management architecture for the NZDPU platform, enabling secure emissions data exchange for 380+ organizations.
- Engineered RESTful APIs, custom user and file management systems, RBAC, and Firebase Authentication across 197+ jurisdictions, scaling to 1,000+ accounts.
- Optimized SQL queries for 2M+ records, reduced manual workflows by 40%, and ensured compliance across 11 industries using GCP.

Tech Stack: Python, JavaScript, SQL, FastAPI, Firebase, GCP

EDUCATION

Saint Joseph's University
BS in Computer Science

May 2025
Philadelphia, PA

LEADERSHIP

Founder & President
Computer Science Club, Saint Joseph's University

2022 - 2025

- Recruited 60+ members and organized workshops, guest lectures, and coding competitions.

PROJECTS

Virtual Travel Platform
Flutter, Python, Firebase, Agora SDK

Cobbs Creek Healthcare

- Built a real-time global travel app with live guides, using Agora for video and Firebase for messaging and authentication.
- Developed cross-platform client/server architecture and designed upcoming LLM-driven personalization features.

Data Chatbot Interface
Python, React, OpenAI API, SQL, LangChain, LangFlow, Plotly

Cobbs Creek Healthcare

- Developed a multi-agent LLM architecture for autonomous SQL query planning, data retrieval, visualization, and explanation.

AI Research Agent & Literature Review Generator
Python, LangChain, Streamlit, Snowflake, AWS EC2

Cobbs Creek Healthcare

- Developed a multi-agent LLM platform for end-to-end literature review workflows including screening, extraction, quality assessment, and report generation.

Net-Zero Data Public Utility (NZDPU) Platform
Python, GCP, PostgreSQL, FastAPI

Insomniac Design

- Implemented backend architecture, custom user and file systems, and optimized SQL for 2M+ emissions records while reducing latency and automating workflows.

TECHNICAL SKILLS

Core Skills: AI/LLM systems, multi-agent workflows, data engineering, full-stack development, cloud analytics

Languages & Tools: Java, C/C++, Python, TypeScript, SQL, JavaScript, AWS, GCP, Snowflake, FastAPI, React, Streamlit, LangChain